

LABUAN, Malaysia

WMO: 964650

Lat: 5.301N

Lon: 115.250E

Elev: 101

StdP: 14.64

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	74.2	74.9	69.9	110.2	83.9	71.2	115.6	82.6	17.4	82.6	16.4	82.7	4.1	240	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.5	91.6	79.8	90.6	79.7	89.8	79.6	82.4	87.9	81.8	87.3	81.2	86.7	5.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	160.7	86.3	80.4	158.2	85.8	79.4	153.2	84.8	46.5	88.4	45.7	87.6	45.1	87.8	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.7	11.2	DB	71.2	94.4	2.3	2.4	69.5	96.2	68.1	97.5	66.8	98.9	65.1	100.6	
			WB	69.8	83.8	2.3	1.4	68.1	84.9	66.7	85.7	65.4	86.5	63.7	87.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	81.6	81.7	82.7	83.6	83.8	83.2	82.8	83.0	82.7	82.2	82.3	82.1
	DBStd	1.96	1.78	1.80	1.68	1.73	1.92	1.88	2.01	2.08	1.86	1.91	1.62	1.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11914	979	887	1014	1008	1047	995	1018	1022	981	999	968	996
	CDD65	6439	514	467	549	558	582	545	553	557	531	534	518	531
	CDH74	71182	5293	4705	5915	6496	6915	6238	6193	6333	5940	5797	5632	5725
	CDH80	23879	1411	1217	1873	2444	2705	2267	2174	2294	2040	1868	1804	1782
Wind	WSAvg	4.4	6.0	6.3	5.2	4.0	3.6	3.7	3.6	3.9	3.9	4.1	4.0	4.7
Precipitation	PrecAvg	129.2	9.8	5.0	6.2	9.2	11.5	11.3	10.8	11.1	12.8	15.1	14.4	11.9
	PrecMax	183.3	29.4	15.1	15.9	18.9	20.2	22.5	23.9	23.9	34.6	26.9	24.4	19.8
	PrecMin	90.9	0.9	1.0	1.6	3.9	5.6	5.2	3.7	3.8	5.1	6.7	8.3	3.0
	PrecStd	21.3	7.0	4.0	3.5	3.9	3.8	4.8	4.6	5.5	5.4	5.9	4.2	4.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.5	90.4	91.6	92.8	92.9	91.7	91.6	91.7	91.5	91.0	90.5	90.0
		MCWB	79.0	79.2	79.2	80.1	80.7	79.8	79.1	79.1	79.2	79.7	80.0	79.6
	2%	DB	87.8	88.1	89.6	90.6	91.1	90.0	89.9	90.0	89.7	89.4	89.1	88.8
		MCWB	79.2	79.1	79.2	80.1	80.5	79.9	79.3	79.1	79.1	79.5	79.9	79.6
	5%	DB	86.4	86.6	88.1	89.5	89.7	89.4	89.0	89.3	88.4	88.0	87.9	87.8
		MCWB	78.8	78.8	79.1	79.9	80.1	79.9	79.4	79.2	79.2	79.4	79.6	79.5
	10%	DB	85.7	85.7	87.0	88.1	88.6	88.0	87.7	87.9	87.6	87.1	86.8	86.3
		MCWB	78.6	78.4	78.8	79.9	80.0	79.8	79.3	79.3	79.3	79.2	79.3	79.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	81.4	81.7	82.6	82.9	82.7	82.1	82.4	82.2	82.1	82.6	81.9
		MCDB	85.9	86.7	87.9	88.2	88.7	88.2	87.6	87.6	87.6	87.5	87.4	86.5
	2%	WB	80.3	80.3	80.7	81.7	82.1	81.8	81.1	81.3	81.0	80.8	81.0	80.8
		MCDB	85.0	85.4	86.4	87.3	87.7	87.4	86.6	86.8	86.6	86.5	86.2	85.9
	5%	WB	79.7	79.6	80.0	81.0	81.4	81.0	80.5	80.6	80.3	80.1	80.3	80.1
		MCDB	84.2	84.3	85.3	86.5	87.0	86.5	86.0	86.2	85.9	85.6	85.5	85.1
	10%	WB	79.1	79.0	79.4	80.4	80.8	80.3	79.9	80.0	79.7	79.5	79.7	79.6
		MCDB	83.4	83.4	84.4	85.8	86.3	85.7	85.2	85.4	85.1	84.8	84.8	84.5
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.0	10.1	10.6	10.5	10.5	10.4	10.5	10.5	10.2	10.1	9.5
		MCDBR	9.5	10.3	11.0	11.5	11.4	11.4	11.2	11.4	11.4	11.1	11.1	10.6
		MCWBR	4.2	4.2	4.2	4.4	4.5	4.8	4.4	4.7	4.8	4.8	4.7	4.5
	5% WB	MCDBR	9.3	9.7	10.5	10.9	11.1	11.1	10.8	11.0	11.0	10.7	10.6	10.2
MCWBR		4.5	4.3	4.4	4.7	4.9	5.2	5.0	5.2	5.3	5.0	4.9	4.6	
Clear-Sky Solar Irradiance	taub	0.410	0.418	0.422	0.424	0.408	0.411	0.409	0.444	0.444	0.444	0.453	0.413	0.409
	taud	2.518	2.468	2.462	2.449	2.510	2.515	2.514	2.388	2.383	2.368	2.502	2.524	
	Ebn at Noon	289	288	286	280	279	275	276	272	276	275	285	287	
	Edh at Noon	34	37	37	37	34	33	33	39	40	40	35	34	
All-Sky Solar Radiation	RadAvg	1544	1754	1867	1899	1744	1635	1633	1690	1649	1551	1564	1507	
	RadStd	141	141	112	82	79	76	78	77	105	111	102	123	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-0.20	N/A	+0.41	+0.61	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.27	N/A	N/A	+0.52	+0.47	+0.41	N/A	N/A	N/A	N/A

Nomenclature: See separate page